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Towards Universal Ultrasound Analysis: Query-driven Multi-Organ Swin-transformer for Joint Segmentation and Classification

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Abstract. Both ultrasound (US) image segmentation and classification are vital for medical image analysis. While deep learning has advanced both tasks, progress is constrained by limited annotated data. Recent foundation models focus mainly on universal segmentation, with far less exploration of universal classification or joint modeling of heterogeneous tasks. We present Query-driven Multi-Organ Swin-transformer (QMOS), a foundation model for universal multi-organ US segmentation and classification. QMOS introduces task-specific branches to capture organ- and task-specific representations, guided by learnable queries for refined feature extraction and cross-task interaction. Evaluated on a multi-organ US dataset, QMOS achieves superior performance in both segmentation and classification tasks and demonstrating robust generalization across diverse organs.

Keywords: Multi-organ Ultrasound, Segmentation and Classification, Query.

1 Introduction

Both ultrasound (US) image segmentation and classification are essential components of medical image analysis, playing a critical role in supporting clinical decision-making [1]. Deep learning has achieved remarkable success in addressing both tasks. However, its application remains limited by the availability of high-quality annotated datasets.

The emergence of the segment anything model (SAM) has accelerated the development of foundation models for medical imaging [2]. Leveraging multi-organ and multi-modal datasets, models such as MedSAM [3], SAMUS [4] and MOFO [5] have been proposed, primarily targeting universal segmentation tasks. Nevertheless, research on universal classification remains relatively scarce, and building foundation models that jointly address heterogeneous segmentation and classification tasks is even more challenging. UniUSNet made an initial attempt by unifying segmentation and classification within a single framework [6].

In this work, we propose Query-driven Multi-Organ Swin-transformer (QMOS) model. QMOS introduces independent task-specific branches for segmentation and classification to store organ- and task-specific representations. Our main contributions are as follows: (a) Our QMOS is a foundation model designed for universal multi-organ

US segmentation and classification. (b) Learnable queries are introduced to refine organ-specific, task-specific representations. (c) Extensive experiments on a multi-organ dataset demonstrate the effectiveness and generalization capability of QMOS.

2 Method

2.1 Model Architectures

The QMOS model adopts a multi-branch transformer-based architecture. It comprises a backbone feature branch, a segmentation-specific branch and a classification-specific branch, as illustrated in Fig. 1.

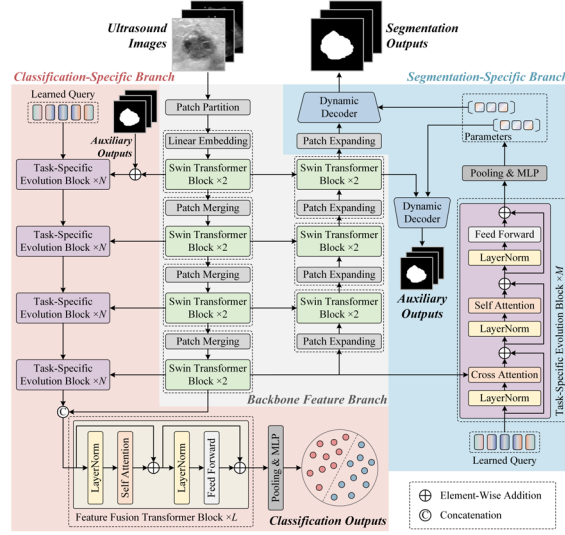


Fig. 1. Overview of the QMOS model

Backbone Feature Branch Given the multi-organ US images as input. A hierarchical Swin-transformer is employed to produce multi-scale organ-invariant representations. These representations are subsequently processed through additional Swin-transformer blocks to reconstruct coarse segmentation representations.

Segmentation-Specific Branch A learnable query is introduced, which interacts with the image representations from the bottleneck layer via task-specific evolution blocks. It extracts organ-specific segmentation representations from the high-dimensional image representations. The resulting representations are subsequently processed to generate two sets of guiding parameters to guide the dynamic decoder to produce the segmentation outputs and auxiliary outputs for deep supervision.

Classification-Specific Branch A learnable query is employed to interact with multi-scale organ-invariant representations through task-specific evolution blocks to produce organ-specific classification representations. Then these representations, together with the image representations from the bottleneck layer, are fed into feature

fusion transformer blocks to integrate cross-task information, followed by a classification head for final prediction.

2.2 Task-Specific Evolution Block

The task-specific evolution block comprises three residual modules: cross-attention, self-attention and feed-forward network. Cross-attention enables the query to interact with image representations, selectively extracting organ-specific information. Subsequently, self-attention and feed-forward network refine these features, thereby strengthening the model's ability to distinguish between organs.

2.3 Dynamic Decoder

A dynamic decoder is proposed to incorporate guiding parameters to reinforce organ-specific anatomical information. Specifically, the dynamic decoder consists of a set of customized convolutions, in which the guiding parameters are employed as the convolutional kernels and biases, to participate directly in the convolution operations.

2.4 Loss Function

Multiple hybrid loss functions are developed to optimize our QMOS model. For both the segmentation outputs and the auxiliary outputs, a combination of Dice coefficient loss and cross-entropy loss are employed to update QMOS, which is formulated as:

$$L_{SEG} = L_{AUX} = 1 - \frac{2 \times \sum_{m=1}^{H \times W} y_m p_m}{\sum_{m=1}^{H \times W} y_m + \sum_{m=1}^{H \times W} p_m} - \sum_{m=1}^{H \times W} [y_m \log(p_m) + (1 - y_m) \log(1 - p_m)] \quad (1)$$

where $y_m \in \{0,1\}$ and $p_m \in [0,1]$ denote the ground truth and the predicted probability. For classification outputs, a weighted cross-entropy loss is utilized:

$$L_{CLS} = -\frac{1}{N} \sum_{n=1}^N [\omega_1 \cdot y_n \cdot \log(p_n) + \omega_0 \cdot (1 - y_n) \cdot \log(1 - p_n)] \quad (2)$$

where $y_n \in \{0,1\}$ and $p_n \in [0,1]$ denote the ground truth and the predicted probability.

2.5 Multi-task Paradigm

The training of QMOS is conducted in two stages: pre-training and cyclic training. In the first stage, QMOS is first trained solely on the segmentation dataset, with all parameters set to be trainable. Then, the parameters of the backbone feature branch and segmentation-specific branch are frozen. Subsequently, QMOS is trained on the classification dataset, updating only the classification-specific branch. In the second stage, cyclic training alternates between segmentation and classification datasets, fine-tuning the pre-trained weights.

3 Experimental Results

3.1 Implementation Details

The QMOS model is implemented in PyTorch 2.0.0, conducted on a NVIDIA RTX 3090 GPU. For performance evaluation, Dice similarity coefficient (DSC) and normalized surface distance (NSD) are adopted for segmentation, while area under the curve (AUC) and accuracy (ACC) are used for classification. A total of 6614 images of five organs are used for segmentation and 4850 images of three organs are employed for classification.

3.2 Results

Segmentation Performance Table 1 presents the segmentation performance of QMOS in comparison with Swin-Unet [7]. QMOS achieves superior results on most organs, with an average DSC of 0.898 and an average NSD of 0.752. The most notable improvement is observed for thyroid, where the DSC increases by 4.1% and the NSD by 5.0% over Swin-Unet.

Table 1. Segmentation performance of Swin-Unet and QMOS.

		Breast	Thyroid	Kidney	Fetus	Heart	Avg.
DSC	Swin-Unet	0.873	0.726	0.905	0.968	0.913	0.877
	QMOS	0.896	0.767	0.925	0.971	0.933	0.898
NSD	Swin-Unet	0.751	0.569	0.763	0.845	0.655	0.717
	QMOS	0.780	0.619	0.798	0.878	0.685	0.752

Classification Performance Table 2 compares the classification performance of QMOS with that of the Swin-transformer [8]. Across most organ categories, QMOS yields higher classification accuracy and discrimination ability, with an average AUC of 0.850 and an average ACC of 0.815.

Table 2. Classification performance of SwinT and QMOS.

		Appendicitis	Liver	Breast	Breast Luminal	Avg.
AUC	SwinT	0.771	0.951	0.914	0.643	0.820
	QMOS	0.798	0.955	0.929	0.718	0.850
ACC	SwinT	0.737	0.929	0.859	0.620	0.786
	QMOS	0.767	0.929	0.872	0.690	0.815

4 Conclusion

We have proposed a foundation model QMOS for universal multi-organ US segmentation and classification. By leveraging task-specific branches with learnable queries, the QMOS effectively captures organ- and task-specific representations, achieving superior performance and strong generalization across diverse organs, and making it a promising solution in future routine clinical practice.

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